EZeats POS Synchronization

Implementation Guide

Backend take-home: POS synchronization

NestJS · PostgreSQL · Drizzle · React

Generated 2026-10-25 from docs/IMPLEMENTATION_GUIDE.md

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1. Project overview and requirements

EZeats is a food-ordering platform. Customers pay for a *check* on their phone. Every paid check must appear on the restaurant's point-of-sale (POS) system so the kitchen can prepare it, and a cancelled check must not leave an open order there.

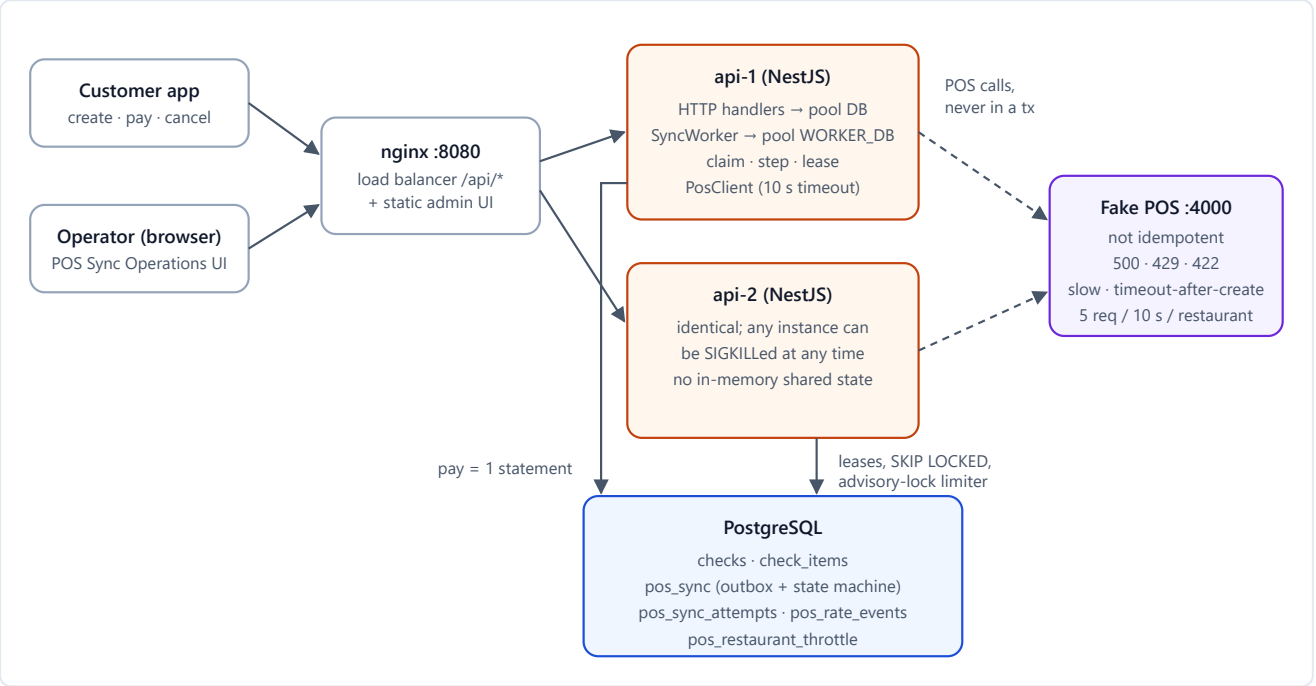
The constraints are what make this hard:

- The API runs as **several identical instances** behind a load balancer. Any instance can be killed at any moment.
- The POS belongs to a third party. It is slow, fails sometimes, rate-limits at **5 requests per 10 seconds per restaurant**, and is **not idempotent**: creating twice with the same `externalRef` creates two orders.

#	Guarantee	How it is met (section)
1	`POST /checks/:id/pay` answers in < 300 ms even if the POS is down or slow	Pay is one SQL statement and never calls the POS (§5, §14)
2	Exactly one POS order per paid check, despite crashes, timeouts, 500/429, many instances	Transactional outbox + leases + `create_sent_at` + externalRef reconciliation (§7, §9)
3	A cancel never reaches the POS before its create; cancelled-before-create leaves nothing	Cancel only for an id the POS returned; suppression before `create_sent_at` (§5, §7)
4	≤ 5 requests / 10 s per restaurant across all instances; a busy restaurant does not delay others	Postgres sliding window + advisory lock; fair claiming (§10)
5	Temporary errors retried; 422 stops, is visible, can be retried by hand	Classified errors, backoff, `failed` state, operator retry (§8, §11)

Required endpoints: `POST /checks`, `POST /checks/:id/pay`, `POST /checks/:id/cancel`, `GET /checks/:id/sync`, `POST /checks/:id/sync/retry`. The fake POS exposes `POST /orders`, `GET /orders?externalRef=`, `POST /orders/:id/cancel` and `GET /debug/orders`.

2. Final architecture



Component diagram

Component	Technology	Responsibility
`apps/api`	NestJS 10, Drizzle ORM, `pg`	HTTP API and the background sync worker. The worker runs inside every instance
PostgreSQL 17	Drizzle schema + generated migrations	The single source of truth: checks, the outbox/state machine, attempt history, rate-limit log
`apps/fake-pos`	NestJS	Third-party POS stand-in with every failure mode from the brief
`apps/admin`	React 18 + Vite	"POS Sync Operations" screen for operators
nginx	`nginx:1.28`	Load-balances `/api/*` across `api-1` and `api-2`, and serves the UI

Instances never talk to each other. Every piece of coordination goes through Postgres row locks, leases and advisory locks.

3. Approaches compared

	A. Synchronous POS call in the request	**B. Postgres outbox + leased workers**	C. Outbox + BullMQ/Redis
Pay latency	Bound by the POS (3–15 s); fails when the POS is down	One statement	One statement + enqueue
Instance killed after POS created	No record → retry duplicates	`create_sent_at` committed before sending; next owner looks the order up	Job may be retried without knowing what the POS did; still needs B's state
Create/cancel ordering	Two requests can race on two instances	One row per check, one lease at a time	Separate jobs can reorder; needs grouping
Shared rate limit	Not shared	Exact, in Postgres	Natural in Redis
Operational cost	Lowest	Postgres only	Two stores and a relay
Breaks when	Always, under any failure	Very high volume (polling, single DB)	Redis/job state disagrees with the DB

4. Why B, and what it costs

All five guarantees hinge on knowing what was committed *before* a POS request left. Option B keeps that knowledge in the same row and transaction as the check, so it cannot disagree with itself.

A queue (C) would add faster wake-ups and a convenient limiter. It would not help with the genuinely hard part, an ambiguous POS outcome, and it would add a second source of truth.

Trade-offs accepted:

- **Polling latency:** up to 250 ms before a paid check is picked up.
- **Load on Postgres:** every step is two short transactions.
- **Postgres as the bottleneck at very high volume** (§16).

5. Request lifecycles

Create `POST /checks` validates the payload with `class-validator` (whitelisted, no extra fields), then inserts the check and its items in one transaction. It answers `201` with `status: "open"`.

Pay `POST /checks/:id/pay` runs one statement:

```
WITH paid AS (
  UPDATE checks SET status='paid', paid_at=now() WHERE id=$1 AND status='open'
  RETURNING id, restaurant_id),
enqueued AS (
  INSERT INTO pos_sync (check_id, restaurant_id, external_ref)
  SELECT id, restaurant_id, id::text FROM paid ON CONFLICT DO NOTHING)
SELECT id FROM paid;
```

- An already-paid check returns `200` unchanged; paying is idempotent.
- A cancelled check returns `409`.
- A concurrent cancel holds the row lock, so the UPDATE re-checks `status='open'` after the cancel commits.

Cancel `POST /checks/:id/cancel` locks the check, marks it cancelled and, if it was paid, sets `cancel_requested` on the sync row in the same transaction. What happens next depends on the row:

- **No worker holds the row:** the state is recomputed. If no create was ever committed as sent, the row closes as `cancelled` immediately and the POS is never contacted.
- **A worker holds the row:** only the flag is set. The worker re-reads it under lock when it finishes its step.

Sync status `GET /checks/:id/sync` returns an explicit DTO (`SyncStatusDto`):

- `state`, and `displayStatus` (processing, retrying, pending, reconciling, cancellation_pending, synced, cancelled, failed, awaiting_payment, not_required)
- `externalRef`, `posOrderId`, `posOrderStatus`
- `attempts`, `consecutiveFailures`, `manualRetries`
- `lastError` with its kind, HTTP status, message and time
- `lastAttemptAt`, `nextAttemptAt`, `processing` (lease owner)
- `createSentAt`, `syncedAt`, `completedAt`
- `retryable` and `retryBlockedReason`

Manual retry `POST /checks/:id/sync/retry` requires the operator token.

- **Sync is `failed`:** the row is locked, and its state is recomputed with `nextAction()`. That is `reconciling` if a create may exist, otherwise `pending`. The failure count resets, and the endpoint answers `202`.
- **Any other state:** it answers `409` with a code and the reason.

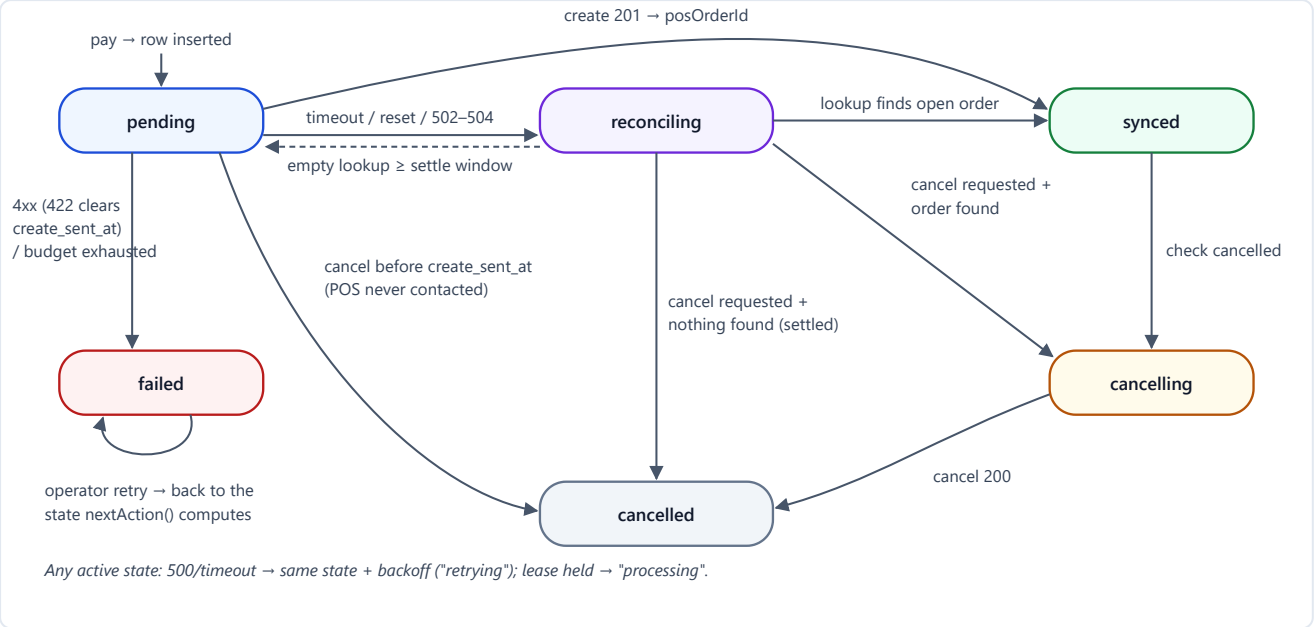
It never calls the POS itself, and five concurrent retries produce one `202` and four `409`s.

6. Database schema

Table	Key columns	Notes
`checks`	`id` uuid PK, `restaurant_id`, `total` numeric(12,2), `status` enum, `paid_at`, `cancelled_at`	CHECK constraint: a cancelled check has `cancelled_at`, a paid one has `paid_at`
`check_items`	`check_id` FK, `position`, `name`, `quantity` > 0, `unit_price`	unique (`check_id`, `position`)
`pos_sync`	`check_id` PK/FK, `external_ref` **unique**, `state` enum, `cancel_requested`, `pos_order_id`, `pos_order_status`, `extra_order_ids[]`, `create_sent_at`, counters, `last_error_*`, `next_attempt_at`, `lease_owner/token/until`	The outbox row and the state machine. Partial index on `next_attempt_at` for active states
`pos_sync_attempts`	`check_id`, `operation`, `ok`, `error_kind`, `http_status`, `message`, `instance_id`, timings	Append-only history shown in the UI
`pos_rate_events`	`restaurant_id`, `sent_at`	Sliding-window log, pruned after 5 minutes
`pos_restaurant_throttle`	`restaurant_id` PK, `blocked_until`, `reason`	Set on an empty budget or a POS 429

`cancel_requested` duplicates `checks.status = 'cancelled'` on purpose. Both are written in the same transaction, and the worker then only needs the row it already holds locked. Migrations are generated by `drizzle-kit`. A drift check (`npm run db:check`) fails when `schema.ts` changes without a migration.

7. Worker lifecycle and state transitions



Sync state machine

Every instance polls every 250 ms. If a cycle found work, it polls again 10 ms later. One cycle:

1. **Claim.** One `UPDATE … FROM (SELECT … FOR UPDATE SKIP LOCKED)` statement leases up to `WORKER_CONCURRENCY` due rows. It takes at most one row per restaurant, skips throttled restaurants, and gives each row a new `lease_token` and `lease_until = now() + 30 s`.
2. **Begin step** (short transaction):
    1. Re-read the row `FOR UPDATE` and verify the lease token.
    2. Compute `nextAction()`.
    3. Take a rate-limit token. If there is none, reschedule and release the row.
    4. For a create, commit `create_sent_at = clock_timestamp()`.
3. **One POS request**, with no transaction open (10 s timeout, redirects not followed).
4. **Finish step** (short transaction):
    1. Record the attempt.
    2. Re-read the *fresh* row `FOR UPDATE` and verify the lease token again. If another instance took over, the result is discarded; that instance's next step reconciles.
    3. Apply `applyOutcome()` and `decideStep()`, persist, and release the lease.

All decisions are pure functions in `sync-machine.ts`, covered by 38 unit tests.

`nextAction()` (cancel before create is impossible by construction):

```
extra duplicate ids?              -> cancel(duplicate)
cancel requested:
   order known & open             -> cancel(order)
   order known & cancelled        -> done: cancelled
   create never sent              -> done: cancelled   (POS never contacted)
   create maybe sent              -> lookup
otherwise:
   order known & open             -> done: synced
   create never sent              -> create
   create maybe sent              -> lookup
```

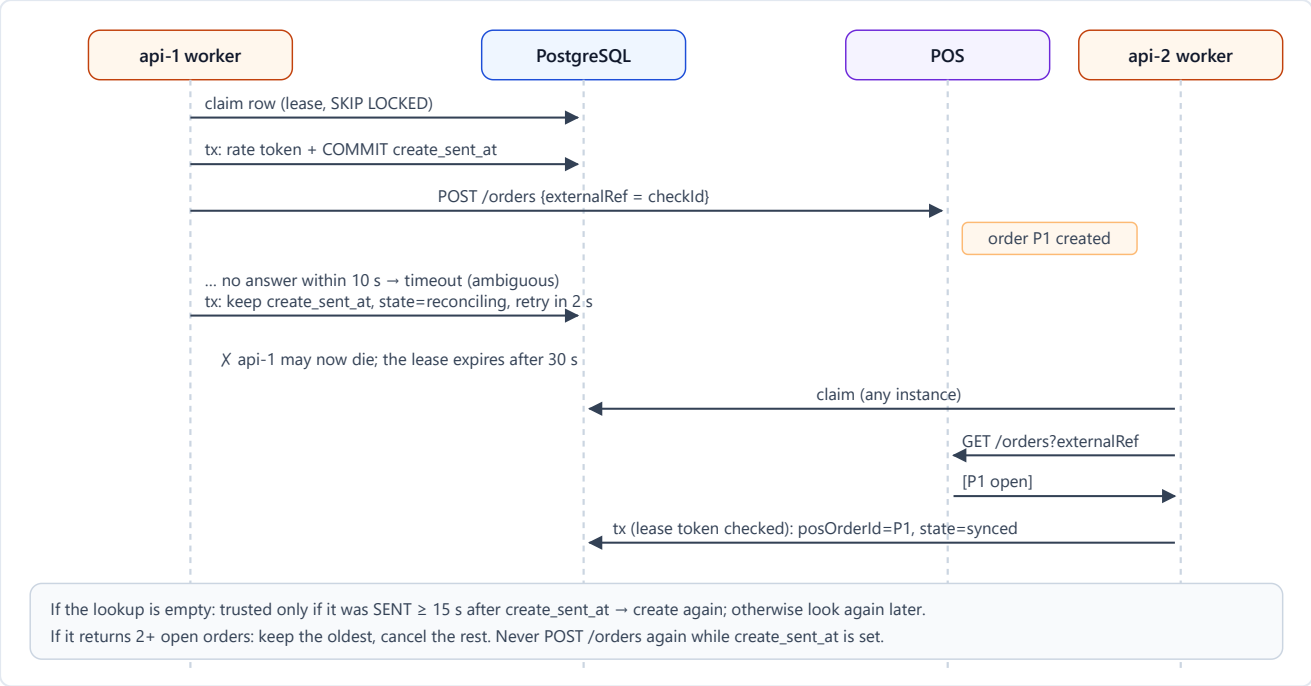
Shutdown. On SIGTERM the worker stops claiming, waits for the claim cycle in progress, then waits for in-flight steps before the connection pools close. A SIGKILL simply leaves leases to expire, after which another instance continues the work.

8. Retry policy

POS outcome	Classification	Create: `create_sent_at`	Next step	Counts toward budget
201 / 200	success	kept	done or next action	resets to 0
500	server_error, not applied	cleared	same action after backoff	yes
429	rate_limited, not applied	cleared	after `Retry-After`; whole restaurant blocked	**no**
422	rejected, **permanent**	cleared	`failed` (operator)	stops
other 4xx	rejected, permanent, **ambiguous**	kept	`failed`; retry or cancel looks up first	stops
timeout, reset, 3xx, 502–504, 2xx without id	ambiguous	kept	lookup in 2 s (`reconciling`)	yes
refused connection	network, not applied	cleared	backoff	yes
lookup or cancel failure	transient	—	repeat (lookup is read-only, cancel is idempotent)	yes

Backoff is exponential with "equal jitter": 1 s × 2^{n-1}, capped at 30 s, half fixed and half random. After 12 consecutive counted failures the row becomes `failed` with `retries_exhausted`. A step whose next action is *done* never parks: for example, a 422 on a check that was cancelled meanwhile simply ends `cancelled`.

9. Exactly-once limits and POS reconciliation



Timed-out create and reconciliation

The POS is not idempotent, so exactly-once is achieved by never creating while a previous create *might* exist.

- **When a create may exist.** `create_sent_at` is committed before every create. While it is set, the only allowed next step is `GET /orders?externalRef=<checkId>`.
- **What the lookup decides:**
  - an open order → adopt it;
  - several open orders → keep the oldest, cancel the rest;
  - nothing → trusted only if the lookup was **sent** at least `SYNC_SETTLE_MS` (15 s) after `create_sent_at`. Otherwise we look again when the window has passed.
- **Timing guards.** `config.ts` refuses to start unless lease ≥ POS timeout + 5 s and settle ≥ POS timeout. That way a second instance cannot reclaim a row while the first one's request may still be in flight.

What we cannot guarantee. If the POS lookup omits an order that exists (eventual consistency), or a create lands more than 15 s after it was sent, a second order can be created. A later lookup would heal it, but none is scheduled once a check is `synced`. No client-side design can close this gap against a non-idempotent POS. An idempotency key or a unique `externalRef` on the POS would.

10. Shared rate limiting

`PosRateLimiter.tryAcquire` runs inside the begin-step transaction:

1. `pg_advisory_xact_lock(hash('pos-rate:' || restaurant))` serializes this restaurant only.
2. If the restaurant has a `pos_restaurant_throttle` row in the future, it is denied.
3. Count `pos_rate_events` in the last 10 s + 0.5 s safety margin. Under 5 → insert an event, and it commits together with the step. Otherwise write a throttle row until the oldest event leaves the window.

A POS `429` writes a throttle row using `Retry-After`, from the database clock, so every instance backs off. The claim query skips throttled restaurants and takes one row per restaurant per cycle. A restaurant with a backlog of 100 checks therefore costs other restaurants nothing.

Integration tests prove it:

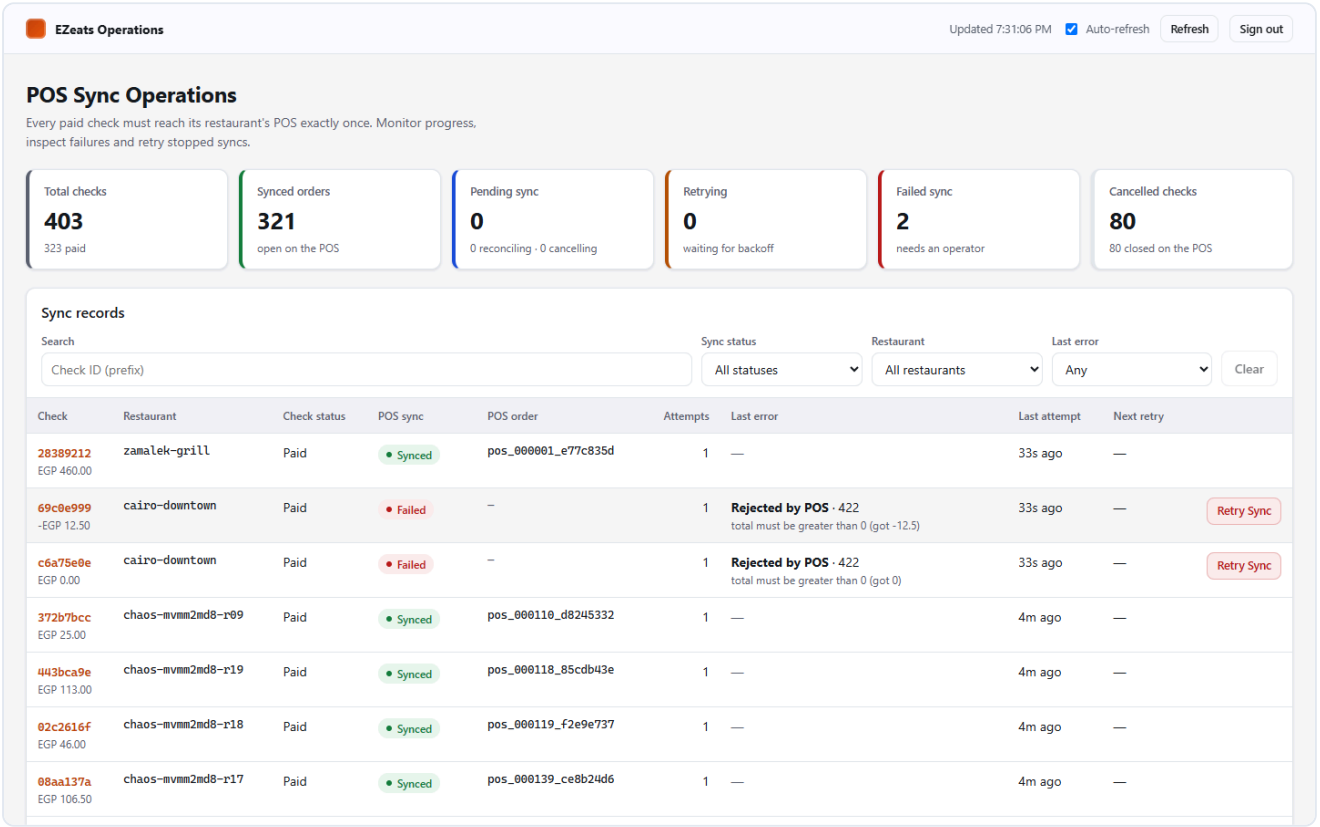
- 20 concurrent acquisitions from two instances yield exactly 5 tokens.
- 12 checks at a busy restaurant span two-plus windows with no 429, while a quiet restaurant syncs within the first window.

11. Operator UI and manual retry

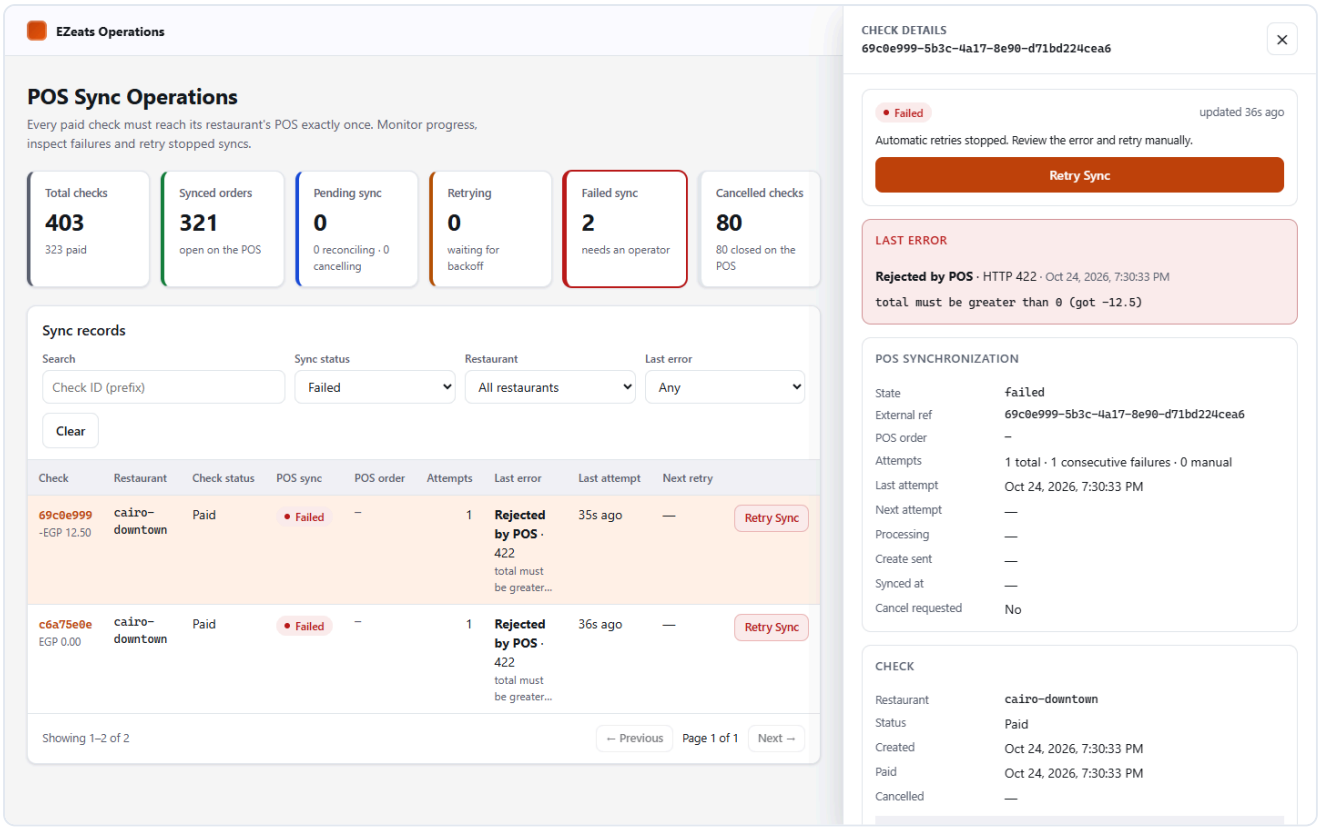
The admin app ("POS Sync Operations") talks only to the real API, through the `/api` proxy.

The screen:

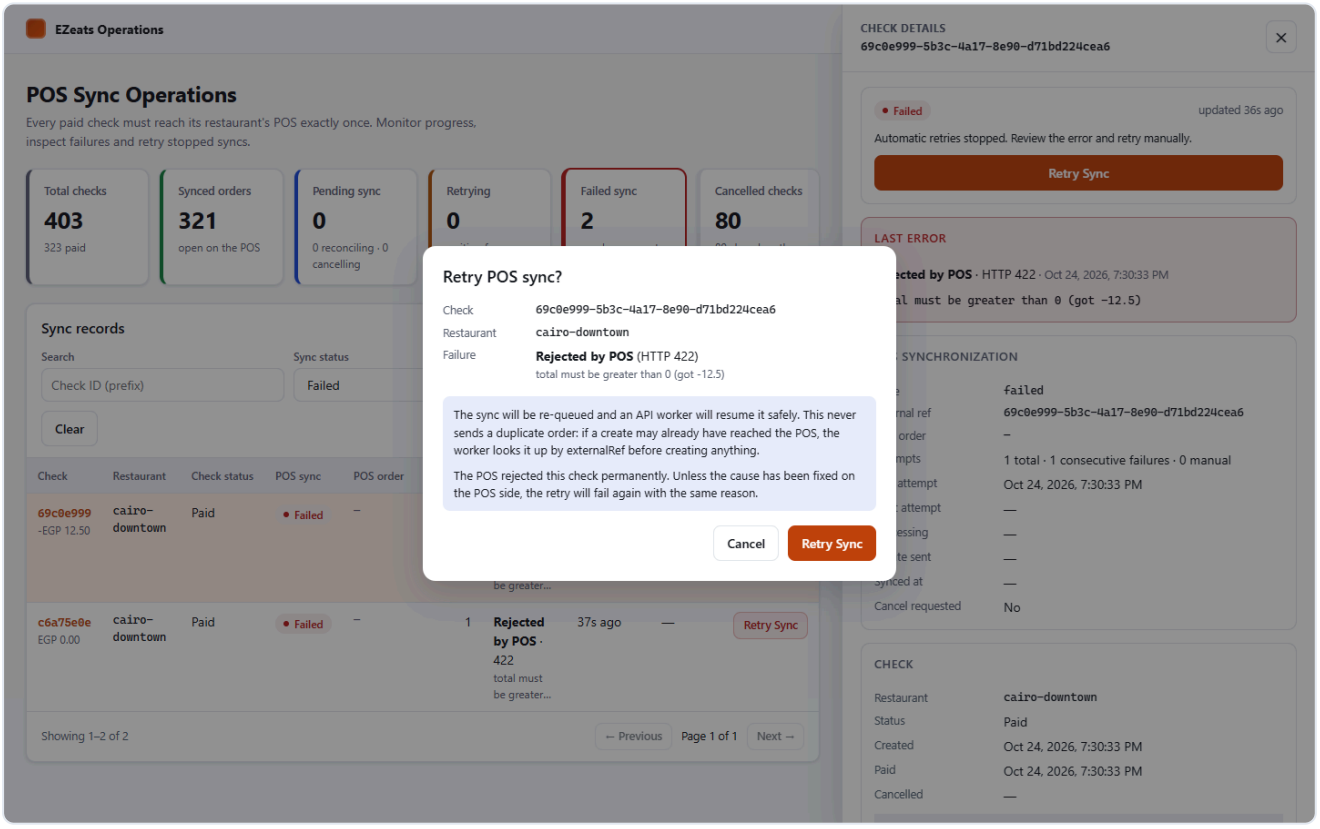
- **Token sign-in.** The token is validated by `GET /admin/session` and kept in session storage.
- **Summary cards.** Total checks, synced, pending, retrying, failed, cancelled. Each card's number equals the rows its filter shows, because the SQL filters mirror `displayStatusOf()`.
- **Searchable, filterable, paginated table.** Filters: status, restaurant, error kind, and check-id prefix.
- **Details drawer.** Check, items and total; sync fields; last error; next attempt; eligibility with its reason; and the persisted POS request history. Nothing is invented: the history comes from `pos_sync_attempts`.
- **Retry Sync.** Shown only when `retryable` is true. A confirmation dialog shows the check id and the failure, and explains that the worker may reconcile before creating. The button is disabled while the request is in flight, and the toast reflects the actual HTTP result, including a `409` reason.
- **Updates.** Polling every 5 s, paused when the tab is hidden, plus a manual refresh. Loading, empty and error states are handled.



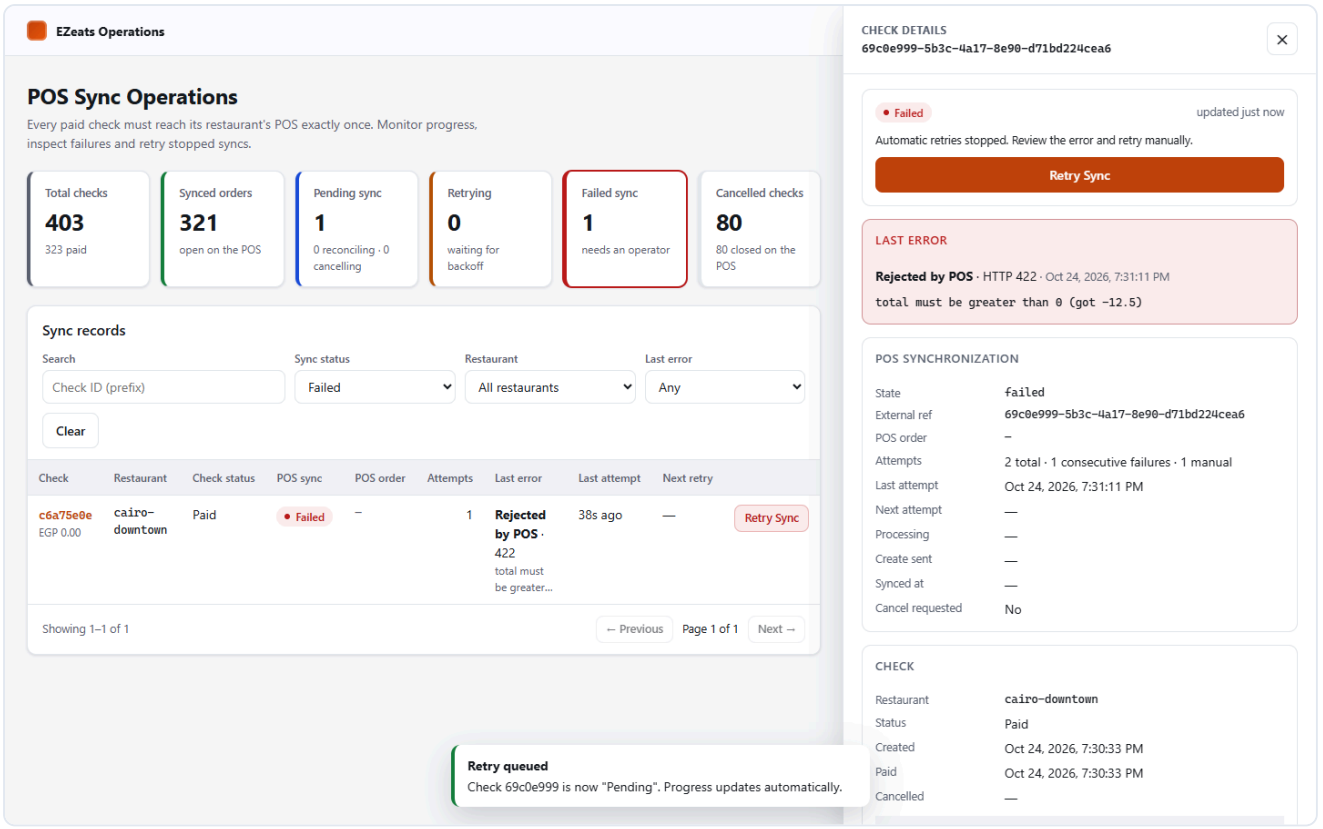
Dashboard



Details drawer



Retry confirmation



Retry result: the 422 failure recurs and stays visible

12. Docker services, environment, local setup

`docker compose up --build` starts:

- `postgres:17-alpine`;
- a one-shot `migrate` job (migrations behind an advisory lock);
- `fake-pos` on port 4000;
- `api-1` and `api-2` on ports 3001 and 3002, with health checks and a 20 s stop grace period;
- `admin` (nginx) on port 8080.

The images are multi-stage builds. Runtime stages contain production dependencies only, run as the `node` user, and use `HUSKY=0`.

The key variables are listed in `README.md`. `.env.example` documents them without secrets, and `.env` is gitignored. Without Docker: `npm ci`, `cp .env.example .env`, `npm run db:migrate`, then start the POS, two API instances and the UI with the `dev:*` scripts.

13. Testing strategy

Layer	Command	What it covers
Unit (94)	`npm run test:unit`	State machine (`nextAction` , `applyOutcome` , `decideStep` , settle window, backoff), POS status classification table, config guards, fake POS store, every Claude Code hook (blocked and allowed inputs)
Integration (37)	`npm run test:int`	Real Postgres + in-process fake POS: create/validation, pay latency with the POS down and slow, cancel open/paid, success, 500 retries, 422 stop, budget exhausted, timeout reconciliation, cancel ordering (before / during / after an ambiguous create), duplicate healing, manual retry (auth, resume, 422, reconcile-first, synced → 409, concurrent), two instances never claiming the same row, crash recovery via lease expiry, stale-lease fencing, shared budget, restaurant isolation, POS 429 handling
Chaos	`npm run test:chaos`	Multi-process, every failure mode, SIGKILL mid-run (§14)

Integration tests run one at a time in a single fork, because they share a database that the global setup rebuilds from the migrations.

14. Chaos test procedure and observed results

Procedure (`tests/chaos/run-chaos.ts`):

1. Start the stack: Docker mode, or local child processes.
2. Wait until no syncs from earlier runs are active, reset the POS, and turn on 20% 500s, 10% timeout-after-create (15 s) and 20% slow responses (3–8 s).
3. Create 200 checks over 20 restaurants and pay all of them through both instances. The test acts as the load balancer and fails over when an instance is down.
4. Cancel 20%: half right after paying (racing the create), half 3 s later.
5. `SIGKILL` `api-1` after 80 payments and restart it 5 s later.
6. Poll until every sync is terminal (10-minute limit), then verify the POS:
    - every kept check has exactly one order, and it is open;
    - every cancelled check has at most one order, and none open;
    - there are no duplicate creates and no orphans.

Observed (local mode, Windows laptop, every process on one machine):

Run	Change since previous	Result	Settled	Pay p50 / p95 / max
1	first run	PASS: 160/160 open, 40 cancelled clean, 0 orphans	62 s	209 / 738 / 1411 ms (incl. failover retries)
2	separate worker pool	PASS	~40 s	128 / 272 / 318 ms
3	review fixes; also fails on any duplicate create	PASS, 0 duplicates	~40 s	114 / 298 / 472 ms

From run 2 on, latency is measured on the successful attempt. The figure including failover to the surviving instance is logged separately. The tail above 300 ms comes from 20 concurrent clients, plus the kill and every service sharing one laptop. It is **reported, not hidden**: under that load the < 300 ms guarantee holds for p95 but not for the maximum. Sequential pay with the POS down or slow stays under 300 ms in the integration tests.

Observed (Docker mode, Docker Desktop on Windows, 2 API containers): four runs in a row passed (160/160 open, 40 cancelled with none open, 0 duplicates, 0 orphans), settling in 36–68 s, with pay p50 / p95 / max of 34–42 / 58–104 / 83–136 ms (successful attempt; up to 7.5 s including failover). An earlier Docker run reported p95 664 ms, but that came from the chaos client blocking its own event loop on a synchronous `docker compose kill` . It now runs docker commands asynchronously.

15. Security and Claude Code setup

Security:

- Operator endpoints (`/admin/*` , retry) require `x-admin-token` , compared in constant time and enforced on the backend. Without `ADMIN_TOKEN` , operator access is denied. This is a development stand-in for real authentication.
- Input is validated and whitelisted; ids are parsed as UUIDs.
- There are no secrets in the repo. The UI never embeds the token.

Claude Code (details in `README.md`):

- `CLAUDE.md` holds the hard rules for the sync.
- `.claude/settings.json` sets the permissions:
  - **allow:** checks, tests and git reads;
  - **ask:** commits, installs, Docker and config edits;
  - **deny:** `.env` , force-push, `reset --hard` , `drizzle-kit push` , `rm -rf` , and editing migrations.
- Four hooks:
  - **command guard** (blocks);
  - **file guard** (blocks);
  - **after-edit check:** Biome, typecheck, migration drift;
  - **Stop gate:** `npm run check` .
- The Husky pre-commit hook runs the same command as the CI `check` job.
- The `sync-reviewer` subagent, run through `/review-sync` , found a real duplicate-order bug (`AI_NOTES.md`).

16. Known limitations and next steps

Limitations:

- Exactly-once depends on the POS lookup being truthful (§9).
- The pay tail latency exceeded 300 ms under the local chaos load (§14).
- A failed cancel of a duplicate parks the whole row for an operator.
- The fake POS keeps its orders in memory.
- Operator auth is a shared token.

At 100× volume:

- Replace polling with `LISTEN/NOTIFY`.
- Partition or archive finished `pos_sync` rows.
- Use a single-row token bucket per restaurant instead of an event log.
- Batch the attempt history.
- Run workers as a separate deployment.
- Add metrics and alerts: age of the oldest due row, failed count, 429 rate.

Next steps:

- Confirm the CI `chaos` job (Docker mode) passes on GitHub; it exists but has not been seen to run.
- Add a verification lookup after sync to catch late duplicates.
- Cut `pay` to one round trip and profile it on separate hosts.
- Add real operator authentication and an audit log.
- Add property-based tests for the state machine.
- Ask the POS vendor for an idempotency key.